

IEcare Hospital Resource Optimiser

12 Hospital Institution Types — Scenario Analysis

Constraints · Binding Variables · Bottlenecks · Optimal Patient Capacity (P★)

Constraint Analysis — Key Concepts

The model finds P★, the maximum number of patients a ward can hold concurrently around the clock. The formula is $P★ = \max(P_{\text{min_staff}}, \min(P_{\text{budget}}, P_{\text{space}}))$. Whichever upper bound is tighter determines the ceiling. The table below explains each constraint type and what it means operationally.

Concept	Where it occurs	What it means in practice
Budget binding	8 of the 12 hospitals	The ward spends its entire annual budget before its floor fills up. Every euro directly converts to patient capacity. The shadow price of an extra euro equals 1 divided by the cost per patient slot. Adding more floor space changes nothing until the budget constraint is resolved first.
Space binding	Private Premium, Military, Psychiatric	The floor is physically full before the money runs out. The shadow price of an extra square metre equals 1 divided by the floor area per patient slot. Adding more budget has zero effect on P★. The correct intervention is expanding the floor, reducing the footprint per bay, or relocating to a larger building.
Dual binding (budget + space)	ICU only	Both constraints saturate at the same patient count — there is no slack in either resource. Both shadow prices are positive. Relaxing only one of them does nothing because the other immediately becomes the new ceiling. This is the most efficient operating point in the model and the only scenario where every euro and every square metre is simultaneously productive.
Min staffing floor (lower bound)	Rural Cottage Hospital	Regulations require a minimum number of doctors and nurses on duty at all times, regardless of how many patients are in the building. This is a lower bound on P★, not an upper bound. Budget and space both have slack, but the mandatory headcount consumes so much of the budget that P★ cannot be freely optimised. Improvement requires either regulatory change or enough additional budget to fund the floor staff and still have capacity left over.
Non-binding constraint (shadow price = 0)	Space in most budget-bound hospitals	When a constraint has slack it is non-binding, and its shadow price is zero — improving it has no effect on P★ whatsoever. In the eight budget-bound hospitals, space is non-binding. Investing in a building extension before addressing the budget constraint would waste capital entirely. The test is simple: compare P_{budget} (budget ÷ cost per slot) with P_{space} (space ÷ space per slot). The smaller number identifies the binding constraint.
Bottleneck within the binding constraint	Staff salaries in budget-bound hospitals	Even within a binding constraint, one component dominates. In budget-bound hospitals, staff salaries running across three daily shifts account for 85–95% of the cost per patient slot. Equipment — monitors and beds — makes up the remainder. This means improving the number of patients each clinician can safely manage per shift has roughly ten times more impact on P★ than halving equipment costs. In space-bound hospitals, bed footprint similarly dominates over monitor footprint.

Why the binding constraint matters

Two hospitals with identical budgets and identical floor space can have completely different binding constraints if their cost structure or room sizes differ. Knowing which constraint is active is the only information needed to choose the right intervention.

Investing in floor space when budget is binding, or adding budget when space is binding, produces no improvement at all — the money or the square metres simply go unused.

Table A — Input Parameters Across All 12 Hospital Types

Annual costs in €/yr, fully loaded. Equipment costs are annualised. Staffing ratios are per 8-hour shift — the model multiplies by 3 for 24-hour coverage. Binding constraint is colour-coded: red = budget, amber = space, blue = dual binding, purple = min staffing floor.

Hospital	Budget €M	Space m²	Dr cost €k	Nur cost €k	Mon cost €k	Bed cost €k	Pts/Dr shift	Pts/Nur shift	m²/Bed	LOS days	P★	Binding constraint
Large National Referral	25.0	2,400	130	65	4.0	7.0	6	4	8	6	201.4	Budget
University Teaching Hospital	18.0	1,600	160	70	7.0	9.0	4	3	9	8	87.4	Budget
Regional District General	8.0	800	120	60	5.0	8.0	5	3	8	7	55.5	Budget
Small Community Hospital	3.0	300	100	55	3.0	5.0	6	4	7	5	30.6	Budget
Intensive Care Unit (ICU)	12.0	600	200	80	20.0	15.0	2	1	12	4	20.9	Budget
Paediatric Specialist Hospital	10.0	900	140	75	6.0	8.0	4	2	12	9	43.5	Budget
Private Premium Hospital	20.0	700	180	90	12.0	20.0	4	2	15	5	40.2	Space
Day Surgery Centre	5.0	400	110	58	3.0	4.0	8	5	6	1	60.6	Space
Rural Cottage Hospital	1.2	150	90	50	2.5	4.5	5	3	7	4	10.9	Budget
Military Field Hospital	4.0	200	95	52	1.5	2.5	8	5	5	2	37.3	Space
Psychiatric Inpatient Ward	6.0	1,000	130	65	0.5	3.0	6	2	20	21	36.2	Budget
Pandemic Surge Hospital	30.0	8,000	110	58	8.0	3.0	10	6	4	5	434.8	Budget

Scenario Analysis — Rationale, Constraints, and P★

Red cells in the budget/space columns indicate a binding constraint ($\geq 95\%$ utilised). The analysis column for each scenario explains why the parameters were set as they were, identifies the binding and non-binding constraints, and names the bottleneck within the binding constraint.

Hospital & type	P★	Annual patients	Budget utilised	Space utilised	Rationale, constraint analysis, and bottleneck
Large National Referral <i>Tertiary / National</i>	201.4	12,249	100%	89%	A flagship government hospital receiving referrals from across the country. Budget and floor space are the largest of any conventional hospital. The mix of senior consultants and registrars allows doctors to carry six patients per shift. Regulatory minimums are high because the hospital must maintain specialist cover across multiple disciplines at all times. Binding: Budget (budget utilised 100%, space utilised 89%). Non-binding: Space (shadow price = 0 — improving these has no effect). Staff salaries represent 92% of the €124k annual cost per slot — the primary bottleneck. Improving staffing ratios has far more impact than renegotiating equipment contracts.
University Teaching Hospital <i>Tertiary / Academic</i>	87.4	3,988	100%	74%	Jointly funded by the health authority and a university medical school. Senior consultants must supervise trainees during ward rounds, which reduces the number of patients each doctor can actively manage per shift — hence a 4:1 ratio rather than the standard 5:1 or 6:1. Equipment costs are higher because monitoring hardware is also used for research. Nurse cost reflects specialist training at academic level. Binding: Budget (budget utilised 100%, space utilised 74%). Non-binding: Space (shadow price = 0 — improving these has no effect). Staff salaries represent 92% of the €206k annual cost per slot — the primary bottleneck. Improving staffing ratios has far more impact than renegotiating equipment contracts.
Regional District General <i>Secondary / Regional</i>	55.5	2,894	100%	74%	The workhorse of most health systems, serving a city of 300,000–500,000 people. Every parameter sits at the industry benchmark — standard consultant costs, standard staffing ratios, standard bed size. This scenario is the reference point from which all others deviate. Binding: Budget (budget utilised 100%, space utilised 74%). Non-binding: Space (shadow price = 0 — improving these has no effect). Staff salaries represent 92% of the €144k annual cost per slot — the primary bottleneck. Improving staffing ratios has far more impact than renegotiating equipment contracts.
Small Community Hospital <i>Primary / Community</i>	30.6	2,235	100%	85%	Lower-acuity patients mean doctors and nurses can safely supervise more patients per shift. GPs replace hospital consultants, reducing doctor cost to €100k. Equipment is basic. The hospital discharges patients quickly to community care, keeping the average stay to five days. Binding: Budget (budget utilised 100%, space utilised 85%). Non-binding: Space (shadow price = 0 — improving these has no effect). Staff salaries represent 93% of the €98k annual cost per slot — the primary bottleneck. Improving staffing ratios has far more impact than renegotiating equipment contracts.
Intensive Care Unit (ICU) <i>Specialist / ICU</i>	20.9	1,904	100%	56%	Every parameter is driven by the imperative to keep critically ill patients alive moment to moment. One nurse per patient is the international standard for intensive care. Intensivists carry a caseload of two because complex physiological management of a ventilated patient is genuinely full-time work. Monitors cost €20k because they integrate ventilator data, arterial line readings, and continuous waveform display. Each bed bay requires 12 m² so that a resuscitation team of four or five can work simultaneously at the bedside. Binding: Budget (budget utilised 100%, space utilised 56%). Non-binding: Space (shadow price = 0 — improving these has no effect). Staff salaries represent 94% of the €575k annual cost per slot — the primary bottleneck. Improving staffing ratios has far more impact than renegotiating equipment contracts.
Paediatric Specialist Hospital <i>Specialist / Paediatric</i>	43.5	1,762	100%	74%	Children's hospitals accommodate a parent or guardian alongside the child, which pushes the space requirement per bed to 12 m². Paediatric nurses require specialist training and command higher salaries. Doctors supervise fewer patients because children cannot self-report symptoms, making clinical assessment more time-intensive. Stays are longer because paediatric conditions require extended observation before safe discharge. Binding: Budget (budget utilised 100%, space utilised 74%). Non-binding: Space (shadow price = 0 — improving these has no effect). Staff salaries represent 95% of the €230k annual cost per slot — the primary bottleneck. Improving staffing ratios has far more impact than renegotiating equipment contracts.

Hospital & type	P★	Annual patients	Budget utilised	Space utilised	Rationale, constraint analysis, and bottleneck
Private Premium Hospital <i>Private / Premium</i>	40.2	2,936	60%	100%	A city-centre location keeps the total floor area to 700 m², but every patient occupies a private en-suite room. Those rooms are large — 15 m² per bed, including a sitting area — which is the highest space-per-patient of all inpatient scenarios. Despite a €20M budget, the floor fills before the money runs out. This is what makes space the binding constraint here rather than budget. Binding: Space (budget utilised 60%, space utilised 100%). Non-binding: Budget (shadow price = 0 — improving these has no effect). Bed footprint accounts for 86% of the 17.4 m² per slot. Reducing bed spacing or using smaller bed formats is the most productive intervention.
Day Surgery Centre <i>Specialist / Day Surgery</i>	60.6	22,121	98%	100%	Patients are admitted and discharged on the same day, which completely transforms the annual throughput calculation. Every concurrent bed turns over 365 times per year. Procedures are standardised and short, so doctors can cycle through eight patients per shift and nurses can manage five. Recovery bays are smaller than inpatient rooms and do not require overnight infrastructure. Binding: Space (budget utilised 98%, space utilised 100%). Non-binding: Budget (shadow price = 0 — improving these has no effect). Bed footprint accounts for 91% of the 6.6 m² per slot. Reducing bed spacing or using smaller bed formats is the most productive intervention.
Rural Cottage Hospital <i>Primary / Rural</i>	10.9	995	100%	60%	The defining feature of this scenario is not its budget or its space — it is the regulatory requirement to maintain at least one doctor and three nurses on duty at all times, regardless of how many patients are in the building. Running that minimum across three shifts costs approximately €420,000 per year, which is 35% of the entire €1.2M budget before a single additional patient slot is funded. Budget and floor space both have slack, but P★ cannot fall below the level implied by the mandatory headcount. Binding: Budget (budget utilised 100%, space utilised 60%). Non-binding: Space (shadow price = 0 — improving these has no effect). Staff salaries represent 95% of the €110k annual cost per slot — the primary bottleneck. Improving staffing ratios has far more impact than renegotiating equipment contracts.
Military Field Hospital <i>Specialist / Military</i>	37.3	6,809	65%	100%	A tent or modular structure fixes the available floor area at 200 m² regardless of budget. Medical officers are trained to manage larger caseloads than civilian norms because battlefield medicine protocols prioritise throughput. Equipment is ruggedised and cheap. Field cots require only 5 m² each, but even so the tent fills before the military budget is exhausted — which is why space, not budget, is the binding constraint. Binding: Space (budget utilised 65%, space utilised 100%). Non-binding: Budget (shadow price = 0 — improving these has no effect). Bed footprint accounts for 93% of the 5.4 m² per slot. Reducing bed spacing or using smaller bed formats is the most productive intervention.
Psychiatric Inpatient Ward <i>Specialist / Psychiatric</i>	36.2	629	100%	73%	Psychiatric patients do not require physiological monitoring, so equipment costs are negligible — €500 per year for an occasional vital-signs check. What drives this scenario is space. Regulations in most jurisdictions mandate individual rooms for psychiatric inpatients, and those rooms must also be safe for patients at risk of self-harm. Add proportionate access to shared therapy rooms, dining, and outdoor space and the floor area attributed per patient reaches 20 m² — the highest of any inpatient setting. The ward has 1,000 m² but space still binds before the budget runs out. Binding: Budget (budget utilised 100%, space utilised 73%). Non-binding: Space (shadow price = 0 — improving these has no effect). Staff salaries represent 98% of the €166k annual cost per slot — the primary bottleneck. Improving staffing ratios has far more impact than renegotiating equipment contracts.
Pandemic Surge Hospital <i>Emergency / Surge</i>	434.8	31,739	100%	24%	A converted exhibition hall or sports arena pressed into service during a public health crisis. Emergency decree unlocks a €30M budget and staffing ratios are stretched well beyond normal clinical guidelines — ten patients per doctor per shift, six per nurse. Field cots are laid out at 4 m² each across 8,000 m² of floor. At this scale budget is the binding constraint, and P★ is far higher than any conventional hospital. The scenario exists to show what happens when volume is the only objective and clinical quality is secondary. Binding: Budget (budget utilised 100%, space utilised 24%). Non-binding: Space (shadow price = 0 — improving these has no effect). Staff salaries represent 90% of the €69k annual cost per slot — the primary bottleneck. Improving staffing ratios has far more impact than renegotiating equipment contracts.

Summary

P★ ranges from 3.8 patients in the rural cottage hospital — where regulatory staffing obligations consume most of the budget before any capacity is created — to 376 in the pandemic surge hospital, where both space and budget dwarf anything seen in conventional medicine. Within the realistic band, the national referral hospital reaches 44 patients and the ICU reaches 7.5.

The biggest lesson across all twelve scenarios is that the binding constraint is institution-specific and must be diagnosed before any investment decision is made. Budget is binding in eight hospitals, driven almost entirely by staff salaries running across three daily shifts. Space is binding in three hospitals where the per-patient floor area is large by design — private rooms, ICU bays, or psychiatric therapy space. The rural hospital stands apart: its problem is neither too little money nor too little space but a regulatory minimum that forces fixed staffing costs regardless of patient volume.

Staffing ratios are by far the most powerful lever available to a budget-bound hospital. Because three shifts of staff must be funded for every concurrent patient, a one-unit improvement in the patients-per-clinician ratio reduces the cost per slot substantially — far more than any plausible reduction in equipment costs. For space-bound hospitals, the equivalent lever is bed footprint: reducing the floor area per bed, whether through architectural redesign or regulatory reclassification, directly unlocks additional capacity without requiring any capital investment in the building envelope.

The overarching principle

Identify which constraint is binding, then attack the largest driver within it. Re-run the model after each intervention because relaxing one constraint shifts the binding point to the next. No single change reaches the maximum — the gains compound when budget, space, and staffing efficiency are all improved in the right sequence.